

Test Plan Summary

ONC 2024 SVAP Syndromic Surveillance Test Plan

Description

null

Test Objectives

The Following test objectives are satisfied in this test plan:

This test plan includes examining a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, an ADT^A03-Discharge message, and an ADT^A01-Admission message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

- Setting (IP, ED, Urgent Care, Ambulatory)
- Chief complaint is captured as a coded value in PV2-3 Admit Reason
- Chief complaint in free-text
- **Diagnoses are submitted test using SNOMED-CT codes** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the ICD-10 CM and SNOMED CT value set (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10 CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case
- **Diagnoses are submitted test using ICD-10-CM codes** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the ICD-10 CM and SNOMED CT value set (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10 CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a pseudonymized manner is required for this Test Step.
- **Patient name is pseudonymized in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a pseudonymized manner is required for this Test Step.
- Multiple races
- Patient's demographic information is unavailable at registration
- Updates to Diagnoses
- Discharge Disposition Home
- Discharge Disposition Expired
- Update Diagnosis Post-Discharge
- Facility sending the syndromic surveillance message is different than the facility where the patient received care
- Sending required observations where known or applicable
 - pregnancy,
 - travel
 - Tobacco use
 - Initial acuity
 - Body Height
 - Body Weight
 - BMI
 - Triage notes
 - Patient Location
 - Chief Complaint
 - Treating Facility Location
 - Age
 - Pediatric case where age is measured in months.
 - Adult case where age is measured in years.
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- Timeliness
- Consistency across messages

Test Case Group: Urgent Care Visit

Description

null

Test Objectives

null

Patient is seen in Urgent Care Center and is discharged/dispositioned to home.

Test Case	SS-UC-1_UC_Visit_Influenza_Child
Description Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 12 hours of any given encounter, and in accordance with local or state laws and practices. The Influenza Child Test Case provides an example of a clinical encounter that could take place in an urgent care setting. This test case also provides an example where the facility sending the syndromic surveillance message is different than the facility where the patient received care. In this test case, an infant is seen at an urgent care center for fever, cough, and earache. Diagnostic tests show that he has influenza and stenosis of the external ear canal. He is discharged to home with treatment instructions and a prescription. Syndromic surveillance data about the visit are transmitted to the local health department. This test case involves 3 steps: • a registration message (ADT^A04), • followed by an update message (ADT^A08), • followed by a discharge message (ADT^A03). Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case. Test Objectives This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance The following test objectives and variants are supported by this test case: • Setting is Urgent Care. • Chief Complaint is captured as free text. • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Updates to Diagnoses • Multiple Races • Insurance is known	
Test Steps	

SS-UC-1.1_Registration_A04	Description A mother brings her 6-month old male infant to NIST Hospital's Urgent Care on February 20, 2025, at 8:30 AM. A clerical assistant registers the patient. She records the patient's name, date of birth, race, ethnicity, residence, insurance information (PlanID123 from NIST INSURANCE 1), and health history. The clerical assistant also records the patient's chief complaint in free-text is, "Fever, cough, and earache." NIST Hospital's Urgent Care is an outpatient facility operated by NIST Hospital. NIST Hospital routinely sends electronic syndromic surveillance data to the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation for all of its facilities within that municipal jurisdiction. At 10:00 AM, NIST Hospital's electronic health record module for syndromic surveillance data assembles and transmits a Registration ADT message about this patient encounter on behalf of the Urgent Care facility as part of an automated process that transmits new or updated content every 15 minutes. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. The dates within the message and across the messages in this test case will be subject to visual inspection to verify data consistencies and the ability to conform to the requirement to send this message within 12 hours of the event triggering this message. Test Objectives This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Urgent Care. • Chief complaint is submitted as free-text. • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Pediatric case where age is measured in months. • Multiple Races • Insurance is known
	Description At 9:00 AM a nurse sees the patient and performs a vital sign assessment, noting that the child currently has a temperature of 101.2 with a productive cough and right ear inflammation and fluid build-up. At 9:10 AM, the physician suspects the patient has influenza with other respiratory manifestations and assigns the patient a working diagnoses of influenza with other respiratory manifestations (ICD-10 CM diagnosis code of Influenza due to unidentified influenza virus with other respiratory manifestations J11.1), and Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear (ICD-10 CM diagnosis code of H61.329) and documents the information in the patient's electronic medical record. At 9:15 AM the physician orders a rapid influenza test, chest x-ray, and a treatment. The patient's vital signs gradually return to normal. A nurse weighs the patient, measures his length, and charts the data in the patient's record. This information is used by the physician when ordering the take-home prescription. NIST Hospital's Urgent Care is an outpatient facility operated by NIST Hospital that routinely sends electronic syndromic surveillance data to the the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 9:15 AM, NIST Hospital's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message with the working diagnoses and the height and weight information for this patient encounter on behalf of the Urgent Care facility as part of an automated process that transmits new or updated content every 15 minutes. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Urgent Care.
- Chief complaint is submitted as free-text.
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Updates to Diagnoses
- Multiple Races
- Insurance is known

SS-UC-1.3_Discharge_A03	Description At 10 AM, the patient's laboratory and radiology tests results are provided to the attending physician. The tests indicate that the infant has influenza and does not have pneumonia. The patient's mother is provided with treatment information and prescriptions for her infant son. The working diagnoses (ICD-10 CM J11.1 Influenza due to unidentified influenza virus with other respiratory manifestations and H61.329 Acquired stenosis of external ear canal secondary to inflammation and infection, unspecified ear) are updated as final diagnoses. At 10:20 AM the patient is discharged home. NIST Hospital's Urgent Care is an outpatient facility operated by NIST Hospital that routinely sends electronic syndromic surveillance data to the the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 10:30 AM on February 20, 2025, NIST Hospital's electronic health record module for syndromic surveillance data assembles and transmits a Discharge ADT message about this patient encounter on behalf of the Urgent Care facility as part of a routine that transmits new or updated content every 15 minutes. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Urgent Care. • Chief complaint is submitted as free-text. • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Updates to Diagnoses • Multiple Races • Insurance is known
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Test Case	SS-UC-2_UC_Visit_Hurricane_Impact

Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 12 hours of any given encounter, and in accordance with local or state laws and practices.

The Hurricane Impact Case provides an example of a clinical encounter that could take place in a freestanding urgent care setting.

In this test case, an adult male seeks care for gastrointestinal symptoms. He had been exposed to flood waters following a recent hurricane, and is experiencing vomiting, diarrhea, nausea, and abdominal pain. He is diagnosed with dehydration and treated with IV fluids, and given an injectable antiemetic to help with vomiting. The clinician determines that further investigation and treatment are better handled in an Emergency Room setting, and refers the patient for Emergency care. The jurisdiction requires that the syndromic surveillance messages are sent in with identifiable demographics. Syndromic surveillance data about the visit are transmitted to the state health department. This Test Case will test using SNOMED-CT Dx codes and full, legal patient name. This test case includes documentation of travel and addition of a final diagnosis post-discharge.

This test case involves 3 steps:

- a registration message (ADT^A04),
- followed by an update message (ADT^A08),
- followed by a discharge message (ADT^A03).

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Urgent Care.
- Chief complaint is captured as a narrative in PV2-3 Admit Reason.
- **Patient name is identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identifiable manner is required for this Test Case.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Adult case where age is measured in years.
- Addition of Height and Weight
- Addition of Diagnoses
- Update Diagnosis Post-Discharge
- Insurance is unknown

Test Steps

SS-UC-
2.1_Registration_A04

Description

62-year-old male walks into NIST Urgent Care on October 20, 2024 at 12:00 PM seeking care for gastrointestinal issues following exposure hurricane-related flood waters. The patient is registered by a clerical assistant who records the patient's name, date of birth, residence information, insurance information (PlanID123 from MyInsuranceCompany), race, ethnicity, smoking status as ex-smoker, and patient's chief complaint, which is "vomiting, diarrhea, nausea, and abdominal pain ". The registrar also documents the patient travel history of 'Spent 4 days helping out in the heavily flooded area of Hurricane City. Returned home 2 days prior to symptoms.'

NIST Urgent Care is an independently operated outpatient facility that routinely sends electronic syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) in accordance with a state regulation. The state regulations permit sending identifiable patient information for the purposes of syndromic surveillance. At 12:15 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Registration ADT message about this patient encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Test Objectives

This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Urgent Care.
- Chief complaint is captured as a narrative in PV2-3 Admit Reason.
- **Patient name is identifiable in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step.
- Adult case where age is measured in years.
- Insurance is unknown

SS-UC-2.2_Discharge_A03	Description At 12:30 PM, a nurse weighs the patient and measures his height and weight, which determines the patient BMI, and at 1pm. sets up an IV. This information is charted in the patient's record to support the physician in ordering any medications. At 5:00 PM, the physician assigns the patient the reason for visit of 'Noninfective gastroenteritis and colitis, unspecified' and documents the information in the patient's electronic medical record. The provider sends a stool sample to the lab for confirmation and documents a working diagnosis of Gastroenteritis (disorder) (SNOMED-CT 25374005) in the patient record, and the patient is discharged home. The provider also documents admitting diagnoses of Dehydration (disorder) (SNOMED-CT 34095006), and Nausea and vomiting (disorder) (SNOMED-CT 16932000). NIST Urgent Care is an independently operated outpatient facility that routinely sends electronic syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) in accordance with state regulation. The state regulations permit sending identifiable patient information for the purposes of syndromic surveillance. in accordance with state regulation. At 6:00 PM on October 20, 2024, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Discharge ADT message about this patient encounter to the state. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Urgent Care. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Patient name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Adult case where age is measured in years. • Addition of Height and Weight • Addition of Diagnoses • Insurance is unknown
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SS-UC-2.3_Update_A08	Description The test results are available 4-days post-discharge confirming that the patient had a cryptosporidium infection. At 2PM, the physician reviews the test results and updates the patient's record with this new Cryptosporidiosis (disorder) diagnosis ((SNOMED-CT 240370009), and the working diagnosis 'Gastroenteritis (disorder)' (SNOMED-CT 25374005) is updated as a final diagnosis. NIST Urgent Care is an independently operated outpatient facility that routinely sends electronic syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) in accordance with state regulation. At 6:00 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message with the working diagnoses and the height and weight information for this patient encounter to the State Public Health Department (NIST PH SS FAC 2). The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Urgent Care. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Patient name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Adult case where age is measured in years. • Update Diagnosis Post-Discharge • Insurance is unknown
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Test Case Group: ED Visit

Description

null

Test Objectives

null

Patient seen in ED and dies.

Test Case	SS-ED-1_ED_Visit_Patient_Dies

Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 12 hours of any given encounter, and in accordance with local or state laws and practices.

The ED Visit Test Case provides an example of an ED visit where the patient's demographic information is unavailable at registration, admit/encounter reason is captured as a coded value using an ICD-10 CM code, a working diagnosis and a final diagnosis are captured using ICD-10 CM codes, and the patient dies. This Test Case will test using a pseudonymized patient record.

In this test case, an unconscious, unknown female patient is brought to an emergency department by ambulance after she sustained critical head and neck injuries due to a collision with a bus while she was riding a bicycle. She is placed on a ventilator, diagnosed as having severe head trauma, and expires from her injuries. Syndromic surveillance data about the visit are transmitted to the state health department.

This test case involves 3 steps:

- a registration message (ADT^A04),
- an update message (ADT^A08), and
- a discharge message (ADT^A03).

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Emergency Department.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Adult case where age is measured in years.
- Updates to patient demographics
- Updates to Diagnoses
- Discharge Disposition Expired
- Insurance is known

Test Steps

SS-ED-
1.1_Registration_A04

Description

An unconscious white female with critical injuries to her head and neck is brought by ambulance to NIST Hospital's Emergency Department at 5:00 PM on July 17, 2024. The paramedics report that the patient turned her bicycle in to a bus moving at 35 mph. The collision threw the cyclist, who was not wearing a helmet, head-first to the ground. The patient's vital signs are stable, but she is on ventilation. Unable to find any identification, the patient is registered without her true name, date of birth, race, ethnicity, insurance information, and health history. Her admit/encounter reason is logged as V13.4XXA (Pedal cycle driver injured in collision with car, pick-up truck or van in traffic accident, initial encounter). No working diagnoses are assigned.

NIST Hospital reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2). At 5:15 PM on July 17, 2024, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A04 Registration message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Test Objectives

This test step examines a HIT Module's ability to create ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Emergency Department.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Patient name is de-identified in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step.
- Adult case where age is measured in years.
- Insurance is known

SS-ED-1.2_Update_A08	Description Physical examination, CT scan of the head and neck (ICD-10 Procedure Code: BW290ZZ Computerized Tomography (CT Scan) of Head and Neck using High Osmolar Contrast) reveal a fracture to the base of the patient's skull and massive intracranial bleeding. While the patient's examination and diagnostic tests are proceeding an emergency room tech uncovers the patient's driver's license among the belongings brought to the ED by the paramedics along with a flight stub showing she traveled from Paris 2 days earlier. At 5:20 PM the patient's medical record is updated with working diagnosis ICD-10 codes S02.112A (Type III occipital condyle fracture, initial encounter for closed fracture) and S06.0X9A (Concussion with loss of consciousness of unspecified duration, initial encounter). Also at this time, the patient's license is used to update the medical record with name and age. NIST Hospital's Emergency Department reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) at least once per day. At 5:30 PM on July 17, 2024, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT 08 Update message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines an HIT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Emergency Department. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Adult case where age is measured in years. • Updates to patient demographics • Addition of Diagnoses and Procedure • Insurance is known
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SS-ED-1.3_Discharge_A03	Description At 5:41 PM, the patient dies. The working diagnoses (ICD-10-CM S02.112A and S06.0X9A) are resolved/inactivated, and the final diagnoses are S06.0X9A (Concussion with loss of consciousness of unspecified duration, initial encounter) and S02.112A (Type III occipital condyle fracture, initial encounter for closed fracture). NIST Hospital's Emergency Department reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) at least once per day. At 5:50, the patient is discharged to the morgue. At 6:00 PM on July 17, 2024, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A03 Discharge message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines an HIT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Emergency Department. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Adult case where age is measured in years. • Discharge Disposition Expired • Insurance is known
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Test Case	SS-ED-2_ED_Visit_Patient_Admitted

Description

A woman arrives to the Southern Midwest Medical Center's Emergency Department in active labor, and delivers in the emergency department at 7:30 AM on February 1, 2025. The newborn has a Congenital diaphragmatic hernia (CDH), and has suffered Meconium aspiration syndrome (MAS). A clerical assistant registers the newborn as a patient. She records the date of birth, race, ethnicity, residence, insurance information, and health history. The baby has not yet been given a name and is subsequently updated once provided by the mother. The clerical assistant also enters the patient's chief complaint as, "Emergency Room birth with Congenital diaphragmatic hernia (CDH) (ICD-10 Q79. 0),". Upon examination, the physician determines the newborn should be admitted to the hospital NICU and put on ECMO to stabilize the baby before performing surgery to repair the CDH (0BQT0ZZ Repair Diaphragm, Open Approach). The newborn is discharged home after recovering from surgery. The Southern Midwest Medical Center reports syndromic surveillance data to the state health department (SHD). At 7:45 AM on February 1, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits a Registration message about this encounter to SHD.

This Test Case provides an example of an ED visit for which the patient's chief complaint is captured as free-text, working diagnosis and final diagnosis are captured with SNOMED CT codes, the patient is discharged from the ED and admitted for inpatient care, and the Admit/Encounter Reason is captured with a SNOMED CT code.

This test case involves 4 steps:

- a registration message (ADT^A04),
- an update message (ADT^A08),
- a discharge message (ADT^A03)
- an admit message (ADT^A01).

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, an ADT^A03-Discharge message, and an ADT^A01-Admission message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Emergency Department with Inpatient Admission.
- Chief complaint is captured as narrative in PV2-3 Admit Reason.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:**Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- **Patient Name is Identifiable in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step.
- Multiple Races.
- Pediatric case where age is measured in months.
- Updates to patient demographics
- Updates to Diagnoses.
- Discharge Disposition Inpatient Admission followed by Home.
- Insurance is unknown

Test Steps

SS-ED-2.1_Registration_A04	Description A woman arrives to the NIST Hospital's Emergency Department in active labor, and delivers in the emergency department at 7:30 AM on February 1, 2025. The newborn has a Congenital diaphragmatic hernia (CDH), which is entered as a working diagnosis, and has suffered Meconium aspiration syndrome (MAS), which is entered as final diagnoses as they were identified upon delivery. At 7:35, a clerical assistant registers the newborn as a patient. She records the patient's date of birth, the height and weight taken at birth, race, ethnicity, residence, insurance information (PlanID123 from MyInsuranceCompany). There is no patient name for the newborn at the time of birth. Southern Midwest Medical Center reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 7:45 AM on February 1, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits a Registration message about this encounter. The placeholder patient name and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements. The following test objectives and variants are supported by this test case: • Setting is Emergency Department with Inpatient Admission. • Chief complaint is captured as free-text. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Patient Name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Multiple Races. • Pediatric case where age is measured in months. • Inclusion of Admitting Diagnoses. • Insurance is unknown
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SS-ED- 2.2_Update_A08	Description At 8am, an abdominal CT (CPT 74150 Computed tomography, abdomen; without contrast material) is taken to determine the extent of CDH. The mother has also selected a baby name, which is updated in the record. The Abdominal scan shows the newborn has Pericardial and diaphragmatic defect syndrome (724068001) which is entered as the final diagnosis. NIST Hospital reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 8:15 AM on February 1, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an Update message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines an HIT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is Emergency Department with Inpatient Admission. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Patient Name is Identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Multiple Races. • Pediatric case where age is measured in months. • Updates to patient demographics • Addition of Diagnoses. • Insurance is unknown
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SS-ED- 2.3_Discharge_A03	Description Upon review of the imaging results, the physician determines the newborn should be admitted to the hospital NICU and put on ECMO to stabilize the baby before performing surgery to repair the CDH (0BQT0ZZ Repair Diaphragm, Open Approach). The patient is quickly discharged from the ED and transported within the facility to the NICU . At 8:25 AM, ED staff complete the patient record and administratively discharge the patient from the ED. The working diagnosis is updated to final diagnosis. The patient's final diagnosis is (Pericardial and diaphragmatic defect syndrome (disorder) (SNOMED CT code 724068001)), NIST Hospital reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 8:30 AM on February 1, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits a Discharge message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines an HIT Module's ability to create an ADT^A03 Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is Emergency Department with Inpatient Admission. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Patient Name is Identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Multiple Races. • Pediatric case where age is measured in months. • Updates to Diagnoses. • Discharge Disposition Inpatient Admission. • Insurance is unknown
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SS-ED-2.4_Admission_A01	Description While the patient is prepared for transport within the hospital for ECMO, the clinical staffers complete an admission record. The Reason for admission is "The Abdominal scan shows the newborn has Pericardial and diaphragmatic defect syndrome.". The Admit Diagnosis is recorded as, "Congenital diaphragmatic hernia (CDH)". NIST Hospital reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 8:35 AM on February 1, 2025, the hospitals electronic health record module for syndromic surveillance data assembles and transmits an Admission message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create ADT^A01 Admission message within the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements for syndromic surveillance. • Setting is Emergency Department with Inpatient Admission. • Chief complaint is captured as narrative in PV2-3 Admit Reason. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Patient Name is Identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Multiple Races. • Pediatric case where age is measured in months. • Addition of Admitting Diagnoses. • Discharge Disposition Home. • Insurance is unknown
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Test Case Group: Inpatient Visit

Description

null

Test Objectives

null

Patient is admitted and discharged from hospital.

Test Case	SS-IP-1_Inpatient_Surgery_Mortality

Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 24 hours of the any given encounter, and in accordance with local or state laws and practices.

The IP Visit-Cardiac Surgery Test Case provides an example of an inpatient surgical visit for a patient with multiple diagnoses. In this scenario, syndromic surveillance data are sent to a state public health agency in accordance with guidelines recommended by the International Society for Disease Surveillance (i.e., once following admission, once following an update, and once following a discharge).

In this test case, a middle aged man is admitted for a pre-scheduled surgical removal of a coronary artery obstruction and insertion of stents, spends one night in the hospital for observation, and is discharged home.

This test case involves 3 steps:

- an admission message (ADT^A01),
- an update message (ADT^A08)
- a discharge message (ADT^A03).

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient name is de-identified in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a pseudonymized manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:**Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Updates to Diagnoses
- Discharge Disposition Expired

Test Steps

SS-IP-1.1_Admission_A01	Description On June 17, 2025 at 7:00 AM, a 51-year old male arrives at NIST Hospital for a scheduled surgical removal of a coronary artery obstruction and insertion of stents (Dilation of 2 Cor Art with Intralum Dev, Open Approach (ICD-10-PCS 02710DZ) to treat his cardiac condition (I25110 Atherosclerotic heart disease of native coronary artery with unstable angina pectoris ICD-10-CM). The patient's name, date of birth, race, ethnicity, residence, and insurance information are captured and the patient is escorted to a pre-operation room in the surgical ward where a nurse captures the patient's chief complaint as fatigue and shortness of breath, inputs the reason for visit as I25.110 (Atherosclerotic heart disease of native coronary artery with unstable angina pectoris), and documents angina pectoris for the patient's admitting diagnosis (I20.9).The patient is then prepared for the procedure, and at 8:30 AM the patient has the procedure. It occurs without incident, and the patient goes to post-operation recovery unit to wake from anesthesia, and receive a post-op examination. At 2:00 PM the patient, who is in good condition, is transported to a cardiac unit for overnight observation. NIST Hospital reports inpatient syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) for all new patient admissions at least once per day. At 8:00 PM on June 17, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A01 Admission message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines a Health IT Module's ability to create ADT^A01 Admission message within PHIN Syndromic Surveillance Messaging Guide's conformance requirements. • Setting is inpatient. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Admitting Diagnoses
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SS-IP-1.2_Update_A08	Description At 9:00 AM the next day, June 18, 2025, the patient is visited by the surgeon and an attending physician who perform a physical to check for signs and symptoms of any post-operation complications. The patient reports that he is experiencing chest pain. The physicians both observe that the patient has signs of cardiac arrhythmia (I49.9). However, the patient suddenly suffers a pulmonary embolism and dies at 12pm. The pulmonary embolism (ICD-10 I26 Pulmonary embolism) is documented as a final diagnosis. NIST Hospital reports syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) at least once per day. At 12:30 PM on June 18, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT 08 Update message about this encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines an HIT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is inpatient. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Addition of Diagnoses • Patient Expired
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SS-IP-1.3_Discharge_A03	Description At 12:45pm, June 18, 2025, the body is moved from the inpatient floor to the hospital morgue. The unit coordinator enters the date/time the body leaves the floor as the discharge date and time. The hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A03 Discharge message about this encounter to the State Public Health Department (NIST PH SS FAC 2). NIST Hospital reports inpatient syndromic surveillance data to the State Public Health Department for all new patient discharges at least once per day. At 8:00 PM on June 18, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A03 Discharge message about this encounter to State Public Health Department (NIST PH SS FAC 2). The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines a Health IT Module's ability to create ADT^A03 Discharge message within PHIN Messaging Guide's conformance requirements. • Setting is inpatient. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Discharge Disposition Expired
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Test Case	SS-IP-2_Inpatient_Visit_Pediatric
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Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For Syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 24 hours of the any given encounter, and in accordance with local or state laws and practices.

The IP – Pediatric Medical Diagnosis Test Case provides an example of an inpatient pediatric medical diagnosis visit and treatment for an infant that presents with respiratory symptoms. In this scenario, Syndromic surveillance data are sent to a state public health agency in accordance with guidelines recommended by the International Society for Disease Surveillance (i.e., once following admission, once for updates, and once following discharge).

In this test case, an 8-week old infant is admitted with severe respiratory symptoms, including runny nose, coughing, sneezing, fever, decreased appetite, wheezing and apnea . The infant is treated with Oxygen Therapy and IV Fluids . After reviewing PCR laboratory results, the pediatrician updates the patient record with a diagnosis of Respiratory Syncytial Virus Infection. Following recovery, the infant is discharged home.

This test case involves 3 steps:

- an admission message (ADT^A01),
- an update message (ADT^A08),
- a discharge message (ADT^A03),

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is identifiable in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:**Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Updates to Diagnoses
- Discharge Disposition Home
- Multiple Races

Test Steps

SS-IP-
2.1_Admission_A01

Description

On February 15, 2025 at 7:00 AM, an 8-week old infant is admitted to NIST Hospital with severe respiratory symptoms, including cough, fever, decreased appetite, wheezing and apnea. The infant's pediatrician has ordered the admission, suspecting the child may have contracted Respiratory Syncytial Virus (RSV). A clerical assistant registers the patient for admission. He records the patient's name, date of birth, race, ethnicity, residence, and insurance information. The triage nurse documents the admitting diagnoses as indicated by the pediatrician as:

56018004^Wheezing (finding)

386661006^Fever (finding)^SCT

724229002^Apnea of infancy (disorder)

700213005^Avoidant restrictive food intake disorder (disorder)

417850002^Respiratory tract congestion and cough (disorder)

The triage nurse also documents travel history indicated by the parents that they had spent 4 days at an amusement park and returned 8 days ago.

NIST Hospital reports inpatient syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) for all new patient admissions at least at least once per day. At 7:15 AM on February 15, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A01 Admission message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A01 Admission message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is Identifiable in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:**Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Admitting Diagnoses
- Multiple Races

SS-IP- 2.2_Update_A08	Description The infant is treated with Oxygen Therapy (Extracorporeal Hyperbaric Oxygenation, Continuous ICD-10-PCS 5A05221) and IV Fluids (Introduce Electrol/Water Bal in Periph Vein, Perc ICD-10-PCS 3E0337Z.) After reviewing PCR laboratory results on February 16, 2025 at 1pm, the pediatrician updates the patient record with a diagnosis of Respiratory Syncytial Virus (RSV) Infection, documenting the updated diagnosis at 1pm. NIST Hospital reports inpatient syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) for all new patient admissions at least once per day. At 1:15 PM on February 16, 2015, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A08 Update message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements • Setting is inpatient. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient Name is Identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Updates to Diagnoses • Multiple Races
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SS-IP-2.3_Discharge_A03	Description Following recovery, the infant is discharged home on February 20, 2025 at 5pm. NIST Hospital reports inpatient syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) for all new patient admissions at least once per day. At 5:15 PM on February 20, 2015, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A03 Discharge message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create ADT^A03 Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is inpatient. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient Name is Identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identified manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Updates to Diagnoses • Discharge Disposition Home • Multiple Races
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Test Case Group: Ambulatory Care Visit

Description

null

Test Objectives

null

Patient is seen in an Ambulatory Care Setting and is discharged/dispositioned to home.

Test Case	SS-AC-1_Visit_AmbulatoryMeasles_Visit_Child

Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 12 hours of any given encounter, and in accordance with local or state laws and practices.

The Ambulatory Visit – PCP New Diagnosis Test Case provides an example of an encounter that takes place in the ambulatory setting. This test case also provides an example where the facility sending the syndromic surveillance message is different than the facility where the patient received care. This Test Case will test using ICD-10 Dx codes and pseudonymized patient name. In this test case, a pediatric patient is seen by his Pediatrician for a rash with a high fever, cough, and watery eyes. The family has recently returned from travel to an amusement park. The pediatrician suspects measles and orders a Real-time RT-PCR to confirm, and provides instruction for home isolation to prevent potential spread. Results received post-discharge are positive for measles, and the patient record is updated with the final diagnosis. Syndromic surveillance data about the visit are transmitted to the local health department.

This test case involves 3 steps:

- a registration message (ADT^A04),
- followed by a discharge message (ADT^A03),
- followed by an update message (ADT^A08)

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Ambulatory.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Update Diagnosis Post-Discharge
- Multiple Races
- Insurance unknown

Test Steps

SS-AC-1.1_Registration_A04	Description A mother brings her 10-month old male infant to see his physician at NIST Primary Care on February 20, 2025, at 8:30 AM. A clerical assistant registers the patient. She records the patient's name, date of birth, race, ethnicity, residence, insurance information and health history. The clerical assistant also records the patient's chief complaint in free-text as, "Patient has a rash, high fever, cough, and watery eyes." The mother also reports that they returned 8 days prior to the visit from travel to an amusement park which the clerk also includes in the record. NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 10:00 AM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Registration ADT message about this patient encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements. • Setting is Ambulatory. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identifiedmanner is required for this Test Step. • Pediatric case where age is measured in months. • Multiple Races • Insurance unknown
SS-AC-1.2_Discharge_A03	Description The pediatrician suspects measles and orders a Real-time RT-PCR to confirm, and provides instruction for home isolation to prevent potential spread. A nurse weighs the patient, measures his length, and charts the data in the patient's record at 8:35. This information may be used by the physician if ordering the take-home prescription. At 8:55 AM, the physician assigns the patient the working diagnoses of measles: (ICD-10 CM diagnosis code of B05.9 Measles without complication), Conjunctivitis (ICD-10 CM diagnosis code of H10.10 Acute atopic conjunctivitis, unspecified eye), Fever (ICD-10 CM diagnosis code of R50.9 Fever, unspecified), and Cough (ICD-10 CM diagnosis code of R05.9 Cough, unspecified) are documented as final in the patient's electronic medical record. Suspecting Measles, the provider takes a nasal swab to send for Real-time RT-PCR (CPT-4 code of 87798 Infectious agent detection by nucleic acid). The provider recommends isolation at home, providing education to the parents (CPT-4 code of 98960 Education and Training for Patient Self-Management). The visit is closed at 9:05 am after the nurse collects the sample to send to the lab. NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 6:00 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Discharge ADT message with height and weight information, as well as the working and final diagnoses for this patient encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Test Objectives

This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

- Setting is Ambulatory.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Add Diagnoses
- Multiple Races
- Insurance unknown

SS-AC-1.3_Update_A08	Description Two days post-visit, the patient's laboratory tests results are provided to the pediatrician. The tests indicate that the patient has tested positive for measles and the physician updates the working diagnosis Measles (disorder) (SNOMED-CT 14189004) to a final diagnosis at 1pm. NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. The state regulations permit sending identifiable patient information for the purposes of syndromic surveillance. in accordance with state regulation. The facility's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message about this patient encounter at 6:00 PM. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity. Test Objectives This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is Ambulatory. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient name is de-identified in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in a de-identifiedmanner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Update Diagnosis Post-Discharge • Multiple Races • Insurance unknown
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Test Case	SS-AC-2_Visit_Ambulatory_Injury_Adult

Description

Syndromic surveillance is public health surveillance that emphasizes the use of near "real-time" health data and statistical tools for disease or hazardous event detection, situation awareness for mass gatherings and public health emergencies, and ad hoc and population health trend analyses. For syndromic surveillance purposes, required data on all clinical encounters must be sent to the jurisdictional public health agency within 12 hours of any given encounter, and in accordance with local or state laws and practices.

The Ambulatory Visit – Medical Condition Test Case provides an example of an encounter that takes place in the ambulatory setting. This test case also provides an example where the facility sending the syndromic surveillance message is different than the facility where the patient received care. This Test Case will test using SNOMED-CT Dx codes and full, legal patient name..

In this test case, an adult patient is seen by his Primary Care Provider for pain and swelling in the arm after a fall. The provider performs an x-ray of the affected area, revealing a fracture. The provider adds a diagnosis to the patient record indicating the fractured arm (ICD-10 CM S52.6- Fracture of lower end of ulna.) The provider applies a splint, and he is discharged home.

Syndromic surveillance data about the visit are transmitted to the local health department.

This test case involves 3 steps:

- a registration message (ADT^A04),
- followed by an update message (ADT^A08),
- followed by a discharge message (ADT^A03)

Messages in this test case will be used in a Manual Test Step under functional requirement testing for validating the HIT capability of populating Visit ID (PV1-19.1) and Visit Time (PV1-44.1) each with the same value for all messages included in this Test Case.

Test Objectives

This test case examines a Health IT Module's ability to create an ADT^A04-Registration message, an ADT^A08-Update message, and an ADT^A03-Discharge message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Ambulatory.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is Identifiable in the submissions:**Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identifiable manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:**Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Adult case where age is measured in years.
- Multiple Races
- Insurance known

Test Steps

SS-AC-2.1_Registration_A04	Description A 38 Year-old adult patient is seen by her Primary Care Provider at NIST Primary Care on March 15, 2025 at 2pm for pain and swelling in the arm following a fall. A clerical assistant registers the patient. She records the patient's name, date of birth, race, ethnicity, residence, insurance information (PlanID123 from MyInsuranceCompany), and health history. The clerical assistant also records the patient's chief complaint in free-text as, "Pain and swelling in the forearm following a fall." NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the State Public Health Department (NIST PH SS FAC 2) syndromic surveillance program in accordance with a state regulation. At 2:30 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Registration ADT message about this patient encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements. • Setting is Ambulatory. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient Name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identifiable manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Multiple Races • Insurance known
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SS-AC- 2.2_Update_A08	Description At 2:35 PM, the nurse collects the patient's height and weight, documenting it in the patient's record. The provider examines the patient's arm, and sends her for an x-ray of the affected area, revealing a fracture. The provider adds a diagnosis to the patient record indicating the fractured arm (ICD-10 CM S52.6- Fracture of lower end of ulna) as a working diagnosis. The facility's electronic health record module for syndromic surveillance data assembles and transmits an Update ADT message about this patient encounter to the State Public Health Department (NIST PH SS FAC 2). The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create an ADT^A08 Update message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is Ambulatory. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient Name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identifiable manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Add working Diagnosis • Multiple Races • Insurance known
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SS-AC-2.3_Discharge_A03	Description The provider reviews the images, and finalized the diagnosis of Stress fracture of left ulna (disorder) (SNOMED-CT 11313841000119108). applies a splint at 3:30, and she is discharged home at 3:45PM. At 4:45 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Discharge ADT message with height and weight information, the splinting procedure as well as the diagnosis for this patient. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions. Test Objectives This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements. • Setting is Ambulatory. • Chief complaint is captured as narrative in PV2-3 Admit Reason • Patient Name is identifiable in the submissions:Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019).The patient name presented in an identifiable manner is required for this Test Step. • Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case. • Pediatric case where age is measured in months. • Multiple Races • Update Diagnosis • Insurance known
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Test Case Group: Business Rules

Description

Test Objectives

null
Using messages generated in the Visit use cases, this Test Group addresses business requirements for Syndromic Surveillance.

Test Case	Consistent ID and Timestamps
Description Test Objectives • Demonstrate the capability to populate Visit ID (PV1-19.1) with the same value for all messages included within each executed this Test Case. • Demonstrate the capability to populate Visit Time (PV1-44.1) with the same value for all messages included within each executed this Test Case. Test Steps	

Consistent Visit ID	Description To reflect the requirement in real-world installations, the HIT Module must demonstrate the capability to populate Visit ID (PV1-19.1) with the same value for all messages included within each test case executed. For each executed Visit Test Case (Urgent Care Visit, ED Visit, Inpatient Visit, Ambulatory Care Visit), the tester will visually inspect each of the messages within that test case to verify that Visit ID (PV1-19.1) is the same in each. Test Objectives Demonstrate the capability to populate Visit ID (PV1-19.1) with the same value for all messages included within each executed this Test Case.
Consistent Visit Time	Description To reflect the requirement in real-world installations, the HIT Module must demonstrate the capability to populate Visit Time (PV1-44.1) with the same value for all messages included within each test case executed. For each executed Visit Test Case (Urgent Care Visit, ED Visit, Inpatient Visit, Ambulatory Care Visit), the tester will visually inspect each of the messages within that test case to verify that Visit Time (PV1-44.1) is the same in each. Test Objectives Demonstrate the capability to populate Visit Time (PV1-44.1) with the same value for all messages included within each executed this Test Case.

Test Case	Timeliness
	Description The system under test provides evidence that the system can be configured to trigger syndromic surveillance messages within 12 hours as required by the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm (Jul 2019). The Tester manually reviews the test evidence provided to evaluate whether the HIT Module has the capability to conform to this IG-specified business requirement. Test Objectives This Manual Test Case requires the Tester to manually evaluate the capability of the HIT Module to submit syndromic surveillance messages within 12 hours of the message trigger event as per the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm (Jul 2019) conformance requirements.
Test Steps	
Demonstrate Timeliness	Description The HIT Developer provides a description of the approach taken to configure the system under test with the ability to transmit syndromic surveillance messages within 12 hours as required by the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm (Jul 2019). The HIT Developer provides evidence of such configuration capability. Evidence may be in the form of e.g. configuration files, installation option screens, installation documentation, etc. The Tester manually reviews the test evidence provided to evaluate whether the HIT Module has the capability to conform to this IG-specified business requirement. Test Objectives This Manual Test Step enables the Tester to manually evaluate the capability of the HIT Module to submit syndromic surveillance messages within 12 hours of the message trigger event as per the HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm (Jul 2019) conformance requirements.

